6.2 Algebra of limits

Given some functions $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$, our goal is to understand how limits of $f \pm g$ might behave, as well as limits of $f \cdot g$ and $\frac{f}{g}$. In general we should exercise caution!

Example 6.1: Let $C \in \mathbb{R}$. Consider the sequences $a_n = n + C$ and $b_n = -n$. Then:

$$\lim_{n\to\infty} a_n = +\infty \qquad \lim_{n\to\infty} b_n = -\infty$$

and

$$\lim_{n\to\infty} (a_n + b_n) = \lim_{n\to\infty} (n + C - n) = \lim_{n\to\infty} C = C.$$

This suggests that $+\infty - \infty = C$ for any C. The analogous example for functions would involve $f(x) = x + C$ and $g(x) = -x$.

Example 6.2: Consider the sequences $a_n = 2n$ and $b_n = -n$. Then:

$$\lim_{n\to\infty} a_n = +\infty \qquad \lim_{n\to\infty} b_n = -\infty$$

and

$$\lim_{n\to\infty} (a_n + b_n) = \lim_{n\to\infty} (2n - n) = \lim_{n\to\infty} n = +\infty.$$

Example 6.3: Consider the sequences $a_n = n$ and $b_n = -2n$. Then:

$$\lim_{n\to\infty} a_n = +\infty \qquad \lim_{n\to\infty} b_n = -\infty$$

and

$$\lim_{n\to\infty} (a_n + b_n) = \lim_{n\to\infty} (n - 2n) = \lim_{n\to\infty} (-n) = -\infty.$$

These examples demonstrate that we should be cautious when taking limits of sums or differences of sequences and functions. In particular, these examples show that $+\infty - \infty$ is a meaningless expression. Similar absurdities can be achieved by multiplication or division of sequences or functions. Let's list these *indeterminate forms*:

Indeterminate forms

The following expressions do not make sense (they are *indeterminate*):

$$+\infty - \infty \qquad -\infty + \infty \qquad \pm\infty \cdot 0 \qquad \frac{\pm\infty}{\pm\infty} \qquad \frac{0}{0}$$

In contrast, here are the meaningful expressions:

Meaningful expressions

The following expressions *do* make sense:

$$\begin{aligned}
+\infty + C &= +\infty \quad (\text{if } C \in \mathbb{R} \cup \{+\infty\}) \\
-\infty + C &= -\infty \quad (\text{if } C \in \{-\infty\} \cup \mathbb{R}) \\
\pm\infty \cdot C &= \pm\infty \quad (\text{if } C \in \mathbb{R}_+ \cup \{+\infty\}) \\
\pm\infty \cdot C &= \mp\infty \quad (\text{if } C \in \{-\infty\} \cup \mathbb{R}_-) \\
\frac{\pm\infty}{C} &= \pm\infty \quad (\text{if } C \in \mathbb{R}_+) \\
\frac{\pm\infty}{C} &= \mp\infty \quad (\text{if } C \in \mathbb{R}_-) \\
\frac{C}{\pm\infty} &= 0 \quad (\text{if } C \in \mathbb{R})
\end{aligned}$$

A more delicate meaningful expression is

$$\frac{C}{0} = \infty \quad (\text{if } C \in \{-\infty\} \cup \mathbb{R} \cup \{+\infty\} \setminus \{0\})$$

However to determine whether it is $+\infty$ or $-\infty$ we need to look at the numerator and the denominator in a neighborhood of the point in question. This could lead to different left and right limits (in the case that the denominator changes sign at the point in question).

Example 6.4: We start with a simple example:

$$f(x) = \frac{1}{x}.$$

At $x = 0$, we have an expression of the form $\frac{C}{0}$. The constant $C = 1 > 0$ in this case, and the denominator changes sign from negative (left of 0) to positive (right of 0). Hence

$$\lim_{x \to 0^-} \frac{1}{x} = -\infty \qquad \text{and} \qquad \lim_{x \to 0^+} \frac{1}{x} = +\infty.$$

Example 6.5: We analyze the rational function

$$f(x) = \frac{x-2}{x^2 - 2x + 1}.$$

Numerator: $x - 2$ is < 0 when $x < 2$, $= 0$ when $x = 2$ and > 0 when $x > 2$.
Denominator: $x^2 - 2x + 1 = (x-1)^2$ is always ≥ 0, and $= 0$ only at $x = 1$.

At $x = 1$ we encounter the situation $\frac{C}{0}$ with $C < 0$. Near $x = 1$, the denominator is > 0, so that both left and right limits must be $-\infty$ and we have:

$$\lim_{x \to 1} \frac{x-2}{x^2 - 2x + 1} = -\infty.$$

The rules above allow us to formulate the following important theorem:

Theorem 6.5 (Algebra of Limits)**:** Let $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$, and let x_0 either be a real number, or $+\infty$ or $-\infty$. Suppose that as $x \to x_0$, both f and g have limits

$$\lim_{x\to x_0} f(x) = \ell_f \qquad \text{and} \qquad \lim_{x\to x_0} g(x) = \ell_g$$

(these limits can be finite or infinite). Then the following equalities hold ***only*** *when the right hand side is a meaningful expression*:

$$\lim_{x\to x_0} (f(x) \pm g(x)) = \ell_f \pm \ell_g$$
$$\lim_{x\to x_0} (f(x) \cdot g(x)) = \ell_f \cdot \ell_g$$
$$\lim_{x\to x_0} \frac{f(x)}{g(x)} = \frac{\ell_f}{\ell_g},$$

where in the last case, $g(x)$ must be nonzero for all x in a deleted neighborhood of x_0, and it is possible that there are different left and right limits (for instance, if g changes sign at x_0).

Proof. We prove just one case to demonstrate the idea of the proof. Let us prove that

$$\lim_{x\to x_0} (f(x) + g(x)) = \ell_f + \ell_g$$

in the case that both ℓ_f and ℓ_g are real numbers (and not $\pm\infty$). That is, we need to show that for any $\varepsilon > 0$ there exists $\delta = \delta(\varepsilon) > 0$ such that for all $0 < |x - x_0| < \delta$, we have that $|(f(x) + g(x)) - (\ell_f + \ell_g)| < \varepsilon$.

Fix $\varepsilon > 0$ and set $\tilde{\varepsilon} = \frac{\varepsilon}{2} > 0$. We start with f: since $\lim_{x\to x_0} f(x) = \ell_f$,

$$\exists \delta_f = \delta_f(\tilde{\varepsilon}) > 0 \text{ such that } \forall 0 < |x - x_0| < \delta_f,\ |f(x) - \ell_f| < \tilde{\varepsilon}.$$

Similarly, for g, using the same $\tilde{\varepsilon}$:

$$\exists \delta_g = \delta_g(\tilde{\varepsilon}) > 0 \text{ such that } \forall 0 < |x - x_0| < \delta_g,\ |g(x) - \ell_g| < \tilde{\varepsilon}.$$

Define $\delta = \min\{\delta_f, \delta_g\}$. Then for all $0 < |x - x_0| < \delta$,

$$\begin{aligned} |(f(x) + g(x)) - (\ell_f + \ell_g)| &= |(f(x) - \ell_f) + (g(x) - \ell_g)| \\ \textit{(triangle inequality)} \qquad &\le |f(x) - \ell_f| + |g(x) - \ell_g| \\ &< \tilde{\varepsilon} + \tilde{\varepsilon} = 2\tilde{\varepsilon} = \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ was arbitrary, we are done: we have shown that for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $0 < |x - x_0| < \delta$, we have $|\ \underbrace{(f(x) + g(x))}_{\text{this is the function}} - \underbrace{(\ell_f + \ell_g)}_{\text{this is the limit}}\ | < \varepsilon$. □

An immediate corollary is this:

Corollary 6.6: If f and g are continuous at $x_0 \in \mathbb{R}$, then so are $f \pm g$, $f \cdot g$ and $\frac{f}{g}$ continuous at x_0 (the last one only if $g(x_0) \neq 0$).

Another corollary is this:

Corollary 6.7: Rational functions are continuous on their domains and polynomials are continuous on $\mathbb{R}$.

6.3 Comparison theorems

Theorem 6.8: Let $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ and let $x_0 \in \{-\infty\} \cup \mathbb{R} \cup \{+\infty\}$. Suppose that

$$\lim_{x\to x_0} f(x) = \ell_f \qquad \text{and} \qquad \lim_{x\to x_0} g(x) = \ell_g.$$

If $f(x) \le g(x)$ on a deleted neighborhood of x_0, then $\ell_f \le \ell_g$.

Proof. We skip the proof here. It can be found in the textbook by Canuto and Tabacco. □

Theorem 6.9 (Squeeze Theorem)**:** Let $f, g, h : \mathbb{R} \to \mathbb{R}$ and let $x_0 \in \{-\infty\} \cup \mathbb{R} \cup \{+\infty\}$. If

$$\lim_{x\to x_0} f(x) = \lim_{x\to x_0} h(x) = \ell \in \mathbb{R} \qquad \text{and} \qquad f \le g \le h \text{ in a deleted neighborhood of } x_0$$

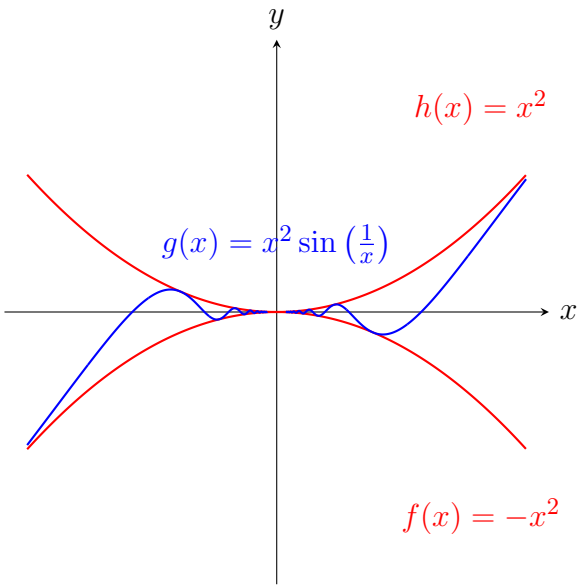
then

$$\lim_{x\to x_0} g(x) = \ell.$$

Proof. We skip the proof here. It can be found in the textbook by Canuto and Tabacco. □

Example 6.6: Using this theorem we can immediately conclude that

$$\lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right) = 0.$$



Example 6.7: The Squeeze Theorem justifies our computation in Example 5.13 (the removable singularity of $\frac{\sin x}{x}$ at $x = 0$). In that problem we found that, for $0 < |x| < \frac{\pi}{2}$,

$$\underbrace{\cos x}_{f(x)} < \underbrace{\frac{\sin x}{x}}_{g(x)} < \underbrace{1}_{h(x)}$$

and used the fact that $\lim_{x\to 0} \cos x = 1$ to conclude that

$$\lim_{x\to 0} \frac{\sin x}{x} = 1.$$

Archimedes squeezes π

Around 250 BC, Archimedes of Syracuse trapped π between two known quantities, in the spirit of the Squeeze Theorem. He drew regular polygons inside and outside a circle: the circle's circumference is larger than the perimeter of the inner polygon and smaller than that of the outer one. Starting from hexagons and repeatedly doubling the number of sides, up to 96, he proved that

$$3.1408\ldots = \frac{223}{71} < \pi < \frac{22}{7} = 3.1428\ldots$$

Archimedes did not have the notion of a limit, but his method contains it: for a circle of radius 1, the inner and outer regular n-gons have perimeters $2n\sin\frac{\pi}{n}$ and $2n\tan\frac{\pi}{n}$, and by the limit $\frac{\sin x}{x} \to 1$ above (together with $\cos x \to 1$), both tend to 2π as $n \to \infty$.

The symbol π itself is much younger. It was introduced in 1706 by the Welsh mathematician William Jones, as the first letter of the Greek word for periphery *(perimeter). It became standard only after Leonhard Euler adopted it in the 1730s, and, through his enormous correspondence, spread it throughout the mathematical world.*

Corollary 6.10 (Squeeze to 0 Theorem)**:** Let $x_0 \in \{-\infty\} \cup \mathbb{R} \cup \{+\infty\}$. Let $f : \mathbb{R} \to \mathbb{R}$ be bounded on a deleted neighborhood of x_0. Let $g : \mathbb{R} \to \mathbb{R}$ satisfy $\lim_{x\to x_0} g(x) = 0$. Then

$$\lim_{x\to x_0} f(x)\cdot g(x) = 0.$$

Proof. Observe that $\lim_{x\to x_0} g(x) = 0$ is satisfied if and only if $\lim_{x\to x_0} |g(x)| = 0$. By assumption, f is bounded on a deleted neighborhood of x_0. This means that there exists $M > 0$ such that $|f(x)| < M$ for all x in this deleted neighborhood. Hence, on this deleted neighborhood of x_0 we have

$$0 \le |f(x)\cdot g(x)| \le M|g(x)|.$$

Since $M|g(x)| \to 0$, Theorem 6.9 gives $\lim_{x\to x_0} |f(x)\cdot g(x)| = 0$, and by the observation at the beginning of the proof (applied to $f\cdot g$), $\lim_{x\to x_0} f(x)\cdot g(x) = 0$. □

Theorem 6.11 (Squeeze to $\pm\infty$ Theorem)**:** Let $f, g : \mathbb{R} \to \mathbb{R}$ and let $x_0 \in \{-\infty\} \cup \mathbb{R} \cup \{+\infty\}$. If

$$\lim_{x\to x_0} f(x) = +\infty \qquad \text{and} \qquad f \le g \text{ in a deleted neighborhood of } x_0$$

then

$$\lim_{x\to x_0} g(x) = +\infty.$$

An analogous statement (with obvious modifications) can be made for the case when the limit is $-\infty$.

Proof. The proof is a simple adaptation of previous proofs, we skip it here. □

Example 6.8: Show that

$$\lim_{x\to-\infty} \left(x^2 - e^x - 3\sin x - 8\right) = +\infty.$$

Observe that for $x < 0$ we have:

$$x^2 - e^x - 3\sin x - 8 \ge x^2 - 1 - 3 - 8 = x^2 - 12.$$

Letting $f(x) = x^2 - 12$, we see that $\lim_{x\to-\infty} f(x) = +\infty$, and the theorem implies the required result.